Adjacent loci are sorted along *different* population axes: recombination time is not the missing ingredient

Module E working note — July 30, 2026

Abstract

The neutral null under-produces the empirical ancestry-sorting signal, and an obvious rescue is that the simulation is simply too young — too few generations of ancestry-breaking recombination. We test this directly. Recombination reduces only the *within*-population component of pooled linkage disequilibrium (LD); the *between*-population component is fixed by the population allele frequencies and is therefore an irreducible floor. Decomposing pooled adjacent-marker LD into these two parts shows that the empirical–null mismatch lives in the between-population floor, which no amount of recombination can change. The direct measurement is the 20-population allele-frequency-vector correlation of *adjacent* diagnostic loci: it is 1.0 for a genome-wide ancestry axis, 0.72 for the drift-only null, and only 0.37 empirically. Empirical populations sort neighbouring loci along **different** axes — population A leans *aquilonia* at one marker, population B at the next — so both loci can have high F_{ST} while their frequencies do not covary and pooled LD stays low. Insufficient recombination is excluded; the residual signal is explicitly locus- (module-) specific sorting.

1 The test

Pooled LD across the 20 populations decomposes into a within-population part and a between-population part:

$$\mathrm{LD}_{\text{pooled}} = \underbrace{\mathrm{LD}_{\text{within}}}_{\text{tract-driven; removed by recombination}} + \underbrace{\mathrm{LD}_{\text{between}}}_{\text{allele-frequency covariance; fixed}} .$$

Recombination is frequency-neutral: it breaks the physical association between neighbouring alleles (the within-population, ancestry-tract part) but leaves every population's allele frequency — and hence the between-population covariance — untouched. So if the simulation is only *too young*, its excess LD should be entirely within-population and should vanish with more recombination. If instead the mismatch is in the between-population floor, recombination cannot help.

We estimate the between-population floor by permuting genotypes among individuals within each population, independently per marker: this destroys within-population LD while holding every population frequency (and F_{ST}) fixed. Because no neutral simulation reproduces the empirical diagnostic $F_{ST} = 0.30$, the "genome-wide ancestry" comparator is a single ancestry axis scaled to that F_{ST} (an α-spread 6.1× the observed), which is the strongest such model. All quantities use diagnostic loci (DI> -25, parental difference > 0.3) and matched population sizes.

2 Result (Fig. 1)

A — tract age. The null's ancestry-LD decays more slowly and lies above empirical at every genetic distance: the null's ancestry tracts are *longer*, i.e. it is younger in recombination terms. On tract length alone, the null would need more recombination. This is the one point in its favour; the decomposition removes it.

B, D — where the LD lives. The null's pooled LD is ∼97% within-population (median r^2: within 0.359, between 0.009) — pure removable tract LD. Empirical LD is 65% within / 35% between (within 0.082, between 0.043), and the between-population floor is *flat with distance*: it is covariance, not linkage. Aging the null removes only the within part, sliding its pooled LD toward 0.009 — *below* empirical (0.127) — while a genome-wide axis at empirical F_{ST} is *all* between-population (0.138), *above* empirical. Neither reaches the empirical value by adding recombination.

Table 1: Within- vs between-population contributions to pooled diagnostic adjacent-marker LD (median r^2). The between-population part is the recombination limit ($t \to \infty$).

	pooled LD	within-pop (removable)	between-pop floor
gen-60 null	0.368	0.359	0.009
empirical	0.127	0.082	0.043
single ancestry axis @ F_{ST}=0.30	0.138	≈0	0.138

C — the decisive measurement. The 20-population frequency-vector correlation of *adjacent* diagnostic loci is 1.00 for a genome-wide axis (all loci track the same α), 0.72 for the drift-only null (weak but ancestry-block structured), and only **0.37** empirically — a broad distribution reaching negative values. Empirical adjacent loci are *less* correlated than even the neutral null: neighbouring diagnostic markers distinguish **different** populations. Two such loci can each have high F_{ST} while their allele frequencies are uncorrelated, so pooled LD stays low — exactly the empirical dissociation.

3 Conclusions

- **Insufficient recombination is excluded.** The empirical–null mismatch is in the between-population floor (Table 1), which recombination cannot change; aging the null overshoots *downward*.
- **A single ancestry axis is excluded.** It forces adjacent-locus correlation to 1.0 and a floor (0.138) above the empirical total LD.
- **Ordinary shared-source and independent-founding histories are excluded.** The independent-founding null already over-correlates adjacent loci ($r = 0.72$); shared ancestry only raises correlation further.
- **An ancient freely-recombining source is hard to defend.** Recombination preserves between-source covariance, so such a source reproduces the high adjacent-locus correlation of a genome-wide axis, not the empirical 0.37. Only an ancient, *structured* metapopulation in which different lineages come to carry the high-frequency allele at adjacent loci could lower it — the demanding scenario left open by the linkage-floor bound.
- **The residual signal is locus- (module-) specific sorting.** Empirical populations sort neighbouring loci along different population axes, producing high F_{ST} without linkage — the signature of divergent, locus-specific selection on ancestry rather than any genome-wide neutral history.

Reproducibility

`dev/R/moduleE_tract_ld.R` (Fig. 1, Table 1); the between-population floor and single-axis construction follow `dev/R/moduleE_ld_floor.R`. This note is the direct, per-locus demonstration behind the linkage-floor bound in `moduleE_complex_demography`.

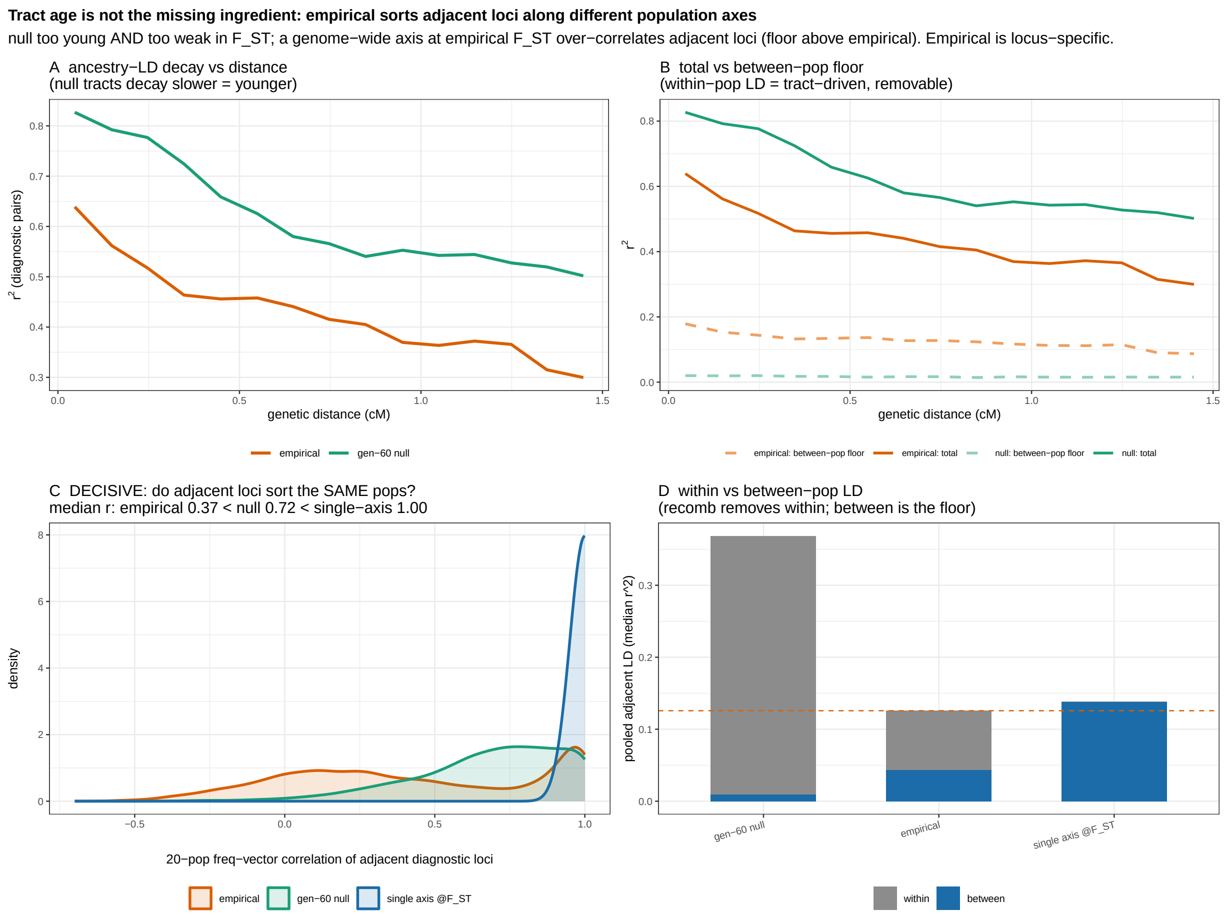

Figure 1: **A** ancestry-LD decay vs genetic distance (null tracts longer = younger). **B** the same decay split into total and the between-population floor (the null's floor is negligible; empirical's is substantial and flat). **C** the 20-population frequency-vector correlation of adjacent diagnostic loci — empirical (median 0.37) is far below the drift-only null (0.72) and a genome-wide axis (1.00). **D** within- vs between-population contributions to pooled adjacent LD; empirical (dashed line) lies between the null floor (0.009, too weak) and the single-axis floor (0.138, too correlated).